

Anany Singh

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EDUCATION

Indraprastha Institute of Information Technology
BTech in Electronics and VLSI Engineering

New Delhi, India
Aug 2023 - Present

EXPERIENCE

Software Development Engineer, Aakaar AI

May 2026 - Present

- Architected and built an AI-powered Listing Agent for Amazon sellers with 2 entry flows (ASIN-based and description-based no-ASIN), generating 7 named slot images and 6 discovery/archetype variations per session, and owned 20 tracked tickets (10 P0) across backend and frontend.
- Designed a 4-table data model (sessions, generations, jobs, generated images) and async job pipeline with status tracking and per-slot versioning, enabling reliable end-to-end generation and syncing of final images back to Amazon listings.
- Implemented free-tier quota enforcement, a 5-ASIN limit and feature gating, then audited and closed 2 quota-bypass paths (ungated no-ASIN flow and `/content/sync`) that allowed unmetered generation.

Undergraduate Researcher, MIDAS Lab

Jan 2026 - Present

- Co-authored a study (arXiv:2607.04537) of 5 code LLMs (GPT-5.3 Codex, Claude Sonnet 4.6, Qwen3-Coder, GLM-5, Kimi K2.5) on 538 executable RunBugRun problems. Confidently wrong debugging instructions cut pass rates by up to 30 points and left every model below its no-guidance baseline.
- Identified “Blind Obedience”: models flagged the wrong instruction as incorrect in most cases, yet still followed it. Confirmed with paired McNemar tests ($p < 0.001$ on all 5 models), with 3.5–13.5× more damage cases than lucky fixes.
- Built a multi-pass agentic evaluation harness (5 repair passes with test-execution feedback) showing that most corrupted programs stay unrecovered, and that higher reasoning budgets did not help (e.g. GLM-5 success fell from 79% to 27%).

PROJECTS

Sputniq AgentOS

May 2026

Tech Stack : *Python, FastAPI, Kafka, Docker* | [GitHub](#)

- Built a config-driven orchestration platform for deploying distributed agentic AI systems, cutting deployment to a single zip upload or one docker compose up. It provisions apps across 15 Docker-in-Docker worker nodes, using a FastAPI control API, a Kafka message bus and a CLI (init, validate, build, package, deploy).
- Enabled scalable deployment of agentic AI workloads across 15 isolated worker nodes, by scripting Docker-in-Docker node provisioning (automatic SSH key generation, image build, and network attachment) and orchestrating the Control API, Kafka and Zookeeper with a single Docker Compose stack.

SignTrack

March 2025

Tech Stack : *Python, OpenCV, Tensorflow, React* | [GitHub](#)

- Developed a real-time computer vision pipeline for traffic-signal detection and counting from dashcam footage, fine-tuning a pretrained YOLOv8 object-detection model via transfer learning on a custom-annotated dataset.
- Built a data augmentation pipeline (brightness/contrast jitter, motion blur, synthetic fog/rain) and tuned confidence thresholds with IoU-based Non-Max Suppression (NMS) to improve generalization and reduce false positives across diverse lighting and weather conditions.
- Geotagged detections using synchronized GPS metadata and reverse geocoding, storing results as spatial points (GeoPandas/GeoJSON) and applying DBSCAN clustering to deduplicate repeated sightings of the same physical signal across frames.

- Designed and implemented a novel Byzantine fault-tolerant clock synchronization protocol that combines the Brooks-Iyengar sensor fusion algorithm with Inter-Tertile Range (ITR) outlier detection to achieve bounded-error time agreement in adversarial peer-to-peer networks, tolerating up to $N/3-1$ malicious nodes.
- Engineered a switch from quartile-based to tertile-based outlier filtering to raise Byzantine tolerance from 25% to the theoretical 33% bound, then applied Brooks-Iyengar interval fusion (originally designed for physical sensor arrays) to aggregate the resulting uncertainty ranges into a single consensus offset with a quantified confidence interval.

RESEARCH PUBLICATIONS

Obey, Diverge, Collapse: Blind Obedience to Incorrect Instructions Drives Code LLMs to Irrecoverable Code Semantic Collapse 2026

arXiv preprint

arXiv:2607.04537

Detection of Fake News

July 2022

International Journal of All Research Education and Scientific Methods (IJARESM)

ISSN: 2455-6211

TECHNICAL SKILLS

Languages and Frameworks: Python, C, C++, Rust, Golang, Java, JavaScript, SQL, React, Node.js, Next.js

Expertise Area: Machine Learning, LLM Evaluation, Agentic Systems, Data Structures, Algorithms, Distributed Systems

Tools and Technologies: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, Git, Docker, Kafka, Google Cloud, Postman

LEADERSHIP / EXTRACURRICULAR

- **Finalist, SEBI Hackathon 2026** (conducted for **Global Fintech Fest**) for **Trust_Rail**, a fintech project [GitHub](#)
- Rated **1397** on **Codeforces**
- Ranked among **Top 500** in the **National Mathematics Talent Contest** with over 60,000 registrations